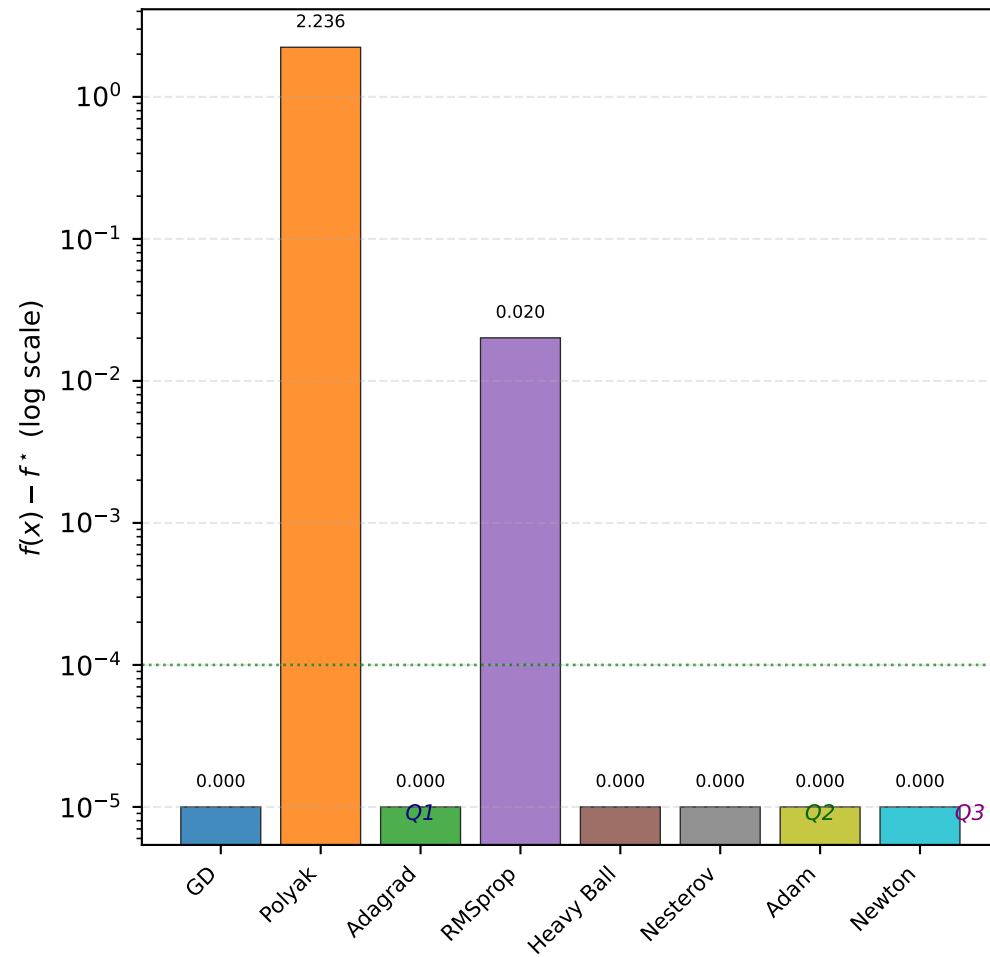
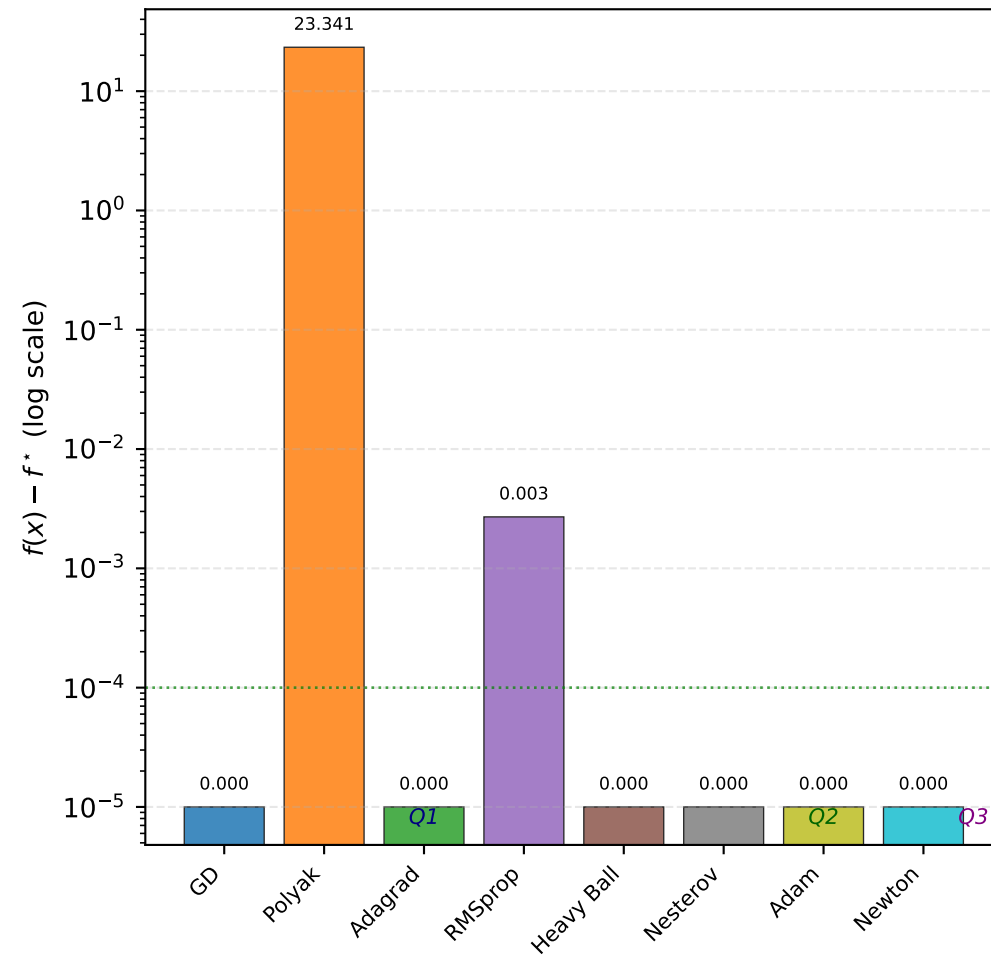


Final Suboptimality $f(x) - f^*$ Across All Methods and Benchmarks
(Polyak uses correct $f^* = 0$; Newton uses 20 iters vs 120-150 for first-order)

Benchmark A ($\kappa \approx 1.12$)



Benchmark B ($\kappa \approx 3.5-7$)



Benchmark C ($\kappa \approx 2500$)

